

PRESCRIBING INFORMATION

CABAZRED 60 mg/1.5 mL Concentrate and Solvent for Solution for Infusion [Cabazitaxel 60 mg/1.5 mL Concentrate and Solvent for Solution for Infusion]

1. Product Name

CABAZRED 60 mg/1.5 mL Concentrate and Solvent for Solution for Infusion.

2. Name and strength of active ingredient(s)

Cabazitaxel 60 mg

3. Product description

Cabazitaxel 60 mg/1.5 mL Concentrate for Solution for Infusion:

Cabazitaxel 60 mg/1.5 mL Concentrate for Solution for Infusion is a clear yellow to brownish-yellow viscous solution free from visible extraneous matter and is available in single use vial containing 60 mg of Cabazitaxel (anhydrous and solvent free)

Excipients: Anhydrous citric acid and Polysorbate 80

Solvent for Cabazitaxel 60 mg/1.5 mL Concentrate for Solution for Infusion:

Solvent for Cabazitaxel 60 mg/1.5 mL Concentrate for Solution for Infusion is a clear colorless solution free from visible extraneous matter containing 13% (w/w) ethanol and Water for Injection q.s.

Cabazitaxel 60 mg/1.5mL Concentrate for Solution for Infusion requires two dilutions prior to intravenous infusion. Cabazitaxel 60 mg/1.5 mL Concentrate for Solution for Infusion should be diluted only with the Solvent for Cabazitaxel 60 mg/1.5mL Concentrate for Solution for Infusion, followed by dilution with either 0.9% sodium chloride solution or 5% dextrose solution.

The reconstituted and diluted solution is clear, slightly yellow solution free from visible extraneous matter.

4. Pharmacodynamics/Pharmacokinetics

Mechanism of action

Cabazitaxel is an antineoplastic agent that acts by disrupting the microtubular network in cells. Cabazitaxel binds to tubulin and promotes the assembly of tubulin into microtubules while simultaneously inhibiting their disassembly. This leads to the stabilisation of microtubules, which results in the inhibition of mitotic and interphase cellular functions.

Pharmacodynamic effects

Cabazitaxel demonstrated a broad spectrum of anti-tumour activity against advanced human tumours xenografted in mice. Cabazitaxel is active in docetaxel-sensitive tumours. In addition,

cabazitaxel demonstrated activity in tumour models insensitive to chemotherapy including docetaxel.

5. Pharmacokinetic properties:

Absorption

Based on the population pharmacokinetic data, after an intravenous dose of cabazitaxel 25 mg/m² every 3 weeks, the C_{max} in patients with metastatic prostate cancer was 226 ng/mL (CV 107%) and was reached at the end of the one-hour infusion (T_{max}). The mean AUC in patients with metastatic prostate cancer was 991 ng•h/mL (CV 34%).

No major deviation to the dose proportionality was observed from 10 to 30 mg/m² in patients with advanced solid tumours.

Distribution

The volume of distribution (V_{ss}) was 4870 L (2640 L/m² for a patient with a median BSA of 1.84 m²) at steady state.

The binding of cabazitaxel to human serum proteins was 89-92% and was not saturable up to 50,000 ng/mL, which covers the maximum concentration. Cabazitaxel is mainly bound to human serum albumin (82.0%) and lipoproteins (87.9% for HDL, 69.8% for LDL, and 55.8% for VLDL). The blood-to-plasma concentration ratios in human blood ranged from 0.90 to 0.99 indicating that cabazitaxel was equally distributed between blood and plasma.

Biotransformation

Cabazitaxel is extensively metabolised in the liver (>95%), mainly by the CYP3A isoenzyme (80% to 90%). Cabazitaxel is the main circulating compound in human plasma. Seven metabolites were detected in plasma (including 3 active metabolites issued from O-demethylations), with the main one accounting for 5% of parent exposure. Around 20 metabolites of cabazitaxel are excreted into human urine and faeces.

The potential risk of inhibition by cabazitaxel at clinically relevant concentrations is possible towards medicinal products that are mainly substrate of CYP3A.

However the cabazitaxel (25 mg/m² administered as a single 1-hour infusion) did not modify the plasma levels of midazolam, a probe substrate of CYP3A. Therefore, at therapeutic doses, co-administration of CYP3A substrates with cabazitaxel to patients is not expected to have any clinical impact.

There is no potential risk of inhibition of medicinal products that are substrates of other CYP enzymes (1A2, 2B6, 2C9, 2C8, 2C19, 2E1, and 2D6) as well as no potential risk of induction by cabazitaxel on medicinal products that are substrates of CYP1A, CYP2C9, and CYP3A. Cabazitaxel did not inhibit the major biotransformation pathway of warfarin into 7-hydroxywarfarin, which is mediated by CYP2C9. Therefore, no pharmacokinetic interaction of cabazitaxel on warfarin is expected.

Cabazitaxel did not inhibit Multidrug-Resistant Proteins (MRP): MRP1 and MRP2 or Organic Cation Transporter (OCT1). Cabazitaxel inhibited the transport of P-glycoprotein (PgP) (digoxin, vinblastine), Breast-Cancer-Resistant-Proteins (BCRP) (methotrexate) and Organic Anion

Transporting Polypeptide OATP1B3 (CCK8) at concentrations at least 15 fold what is observed in clinical setting while it inhibited the transport of OATP1B1 (estradiol-17 β -glucuronide) at concentrations only 5 fold what is observed in clinical setting. Therefore the risk of interaction with substrates of MRP, OCT1, PgP, BCRP and OATP1B3 is unlikely at the dose of 25 mg/m². The risk of interaction with OATP1B1 transporter is possible, notably during the infusion duration (1 hour) and up to 20 minutes after the end of the infusion.

Elimination

After a 1-hour intravenous infusion [¹⁴C]-cabazitaxel at 25 mg/m² in patients, approximately 80% of the administered dose was eliminated within 2 weeks. Cabazitaxel is mainly excreted in the faeces as numerous metabolites (76% of the dose); while renal excretion of cabazitaxel and metabolites account for less than 4% of the dose (2.3% as unchanged medicinal product in urine). Cabazitaxel had a high plasma clearance of 48.5 l/h (26.4 l/h/m² for a patient with a median BSA of 1.84 m²) and a long terminal half-life of 95 hours.

Special populations

Elderly patients

In the population pharmacokinetic data patients of 65 years and older, have no age effect on the pharmacokinetics of cabazitaxel.

Paediatric patients

Safety and effectiveness of cabazitaxel have not been established in children and adolescents below 18 years of age.

Hepatic impairment

Cabazitaxel is eliminated primarily via liver metabolism.

Patients with hepatic impairment showed no influence of mild (total bilirubin >1 to $\leq$ 1.5 x ULN or AST >1.5 x ULN) or moderate (total bilirubin >1.5 to $\leq$ 3.0 x ULN) hepatic impairment on cabazitaxel pharmacokinetics. The maximum tolerated dose (MTD) of cabazitaxel was 20 and 15 mg/m², respectively.

In patients with severe hepatic impairment (total bilirubin >3 ULN), a 39% decrease in clearance was observed when compared to patients with mild hepatic impairment, indicating some effect of severe hepatic impairment on cabazitaxel pharmacokinetics. The MTD of cabazitaxel in patients with severe hepatic impairment was not established.

Based on safety and tolerability data, cabazitaxel dose should be reduced in patients with mild hepatic impairment. Cabazitaxel is contraindicated in patients with severe hepatic impairment.

Renal impairment

Cabazitaxel is minimally excreted via the kidney (2.3% of the dose). Mild to moderate renal impairment did not have meaningful effects on the pharmacokinetics of cabazitaxel.

6. Therapeutic Indications

Cabazitaxel in combination with prednisone or prednisolone is indicated for the treatment of adult patients with hormone refractory metastatic prostate cancer previously treated with a docetaxel-containing regimen.

7. Dosage and method of administration:

The use of cabazitaxel should be confined to units specialised in the administration of cytotoxics and it should only be administered under the supervision of a physician experienced in the use of anticancer chemotherapy. Facilities and equipment for the treatment of serious hypersensitivity reactions like hypotension and bronchospasm must be available (see section Special Warnings and Precautions for Use).

Premedication

The recommended premedication regimen should be performed at least 30 minutes prior to each administration of cabazitaxel with the following intravenous medicinal product to mitigate the risk and severity of hypersensitivity:

- Antihistamine (dexchlorpheniramine 5 mg or diphenhydramine 25 mg or equivalent),
- Corticosteroid (dexamethasone 8 mg or equivalent), and with
- H₂ antagonist (ranitidine or equivalent) (see section Special Warnings and Precautions for Use).

Antiemetic prophylaxis is recommended and can be given orally or intravenously as needed.

Throughout the treatment, adequate hydration of the patient needs to be ensured, in order to prevent complications like renal failure.

Posology

The recommended dose of cabazitaxel is 25 mg/m² administered as a 1 hour intravenous infusion every 3 weeks in combination with oral prednisone or prednisolone 10 mg administered daily throughout treatment.

Dose Adjustments

Dose modifications should be made if patients experience the following adverse reactions (Grades refer to Common Terminology Criteria for Adverse Events (CTCAE 4.0)):

Table 1: Recommended dose modifications for adverse reaction in patients treated with cabazitaxel

Adverse Reactions	Dose Modification
Prolonged grade ≥ 3 neutropenia (longer than 1 week) despite appropriate treatment including G-CSF	Delay treatment until neutrophil count is $> 1,500$ cells/mm ³ , then reduce cabazitaxel dose from 25 mg/m ² to 20 mg/m ² .
Febrile neutropenia or neutropenic infection	Delay treatment until improvement or resolution, and until neutrophil count is $> 1,500$ cells/mm ³ , then reduce cabazitaxel dose from 25 mg/m ² to 20 mg/m ² .
Grade ≥ 3 diarrhoea or persisting diarrhoea despite appropriate treatment, including fluid and electrolytes replacement	Delay treatment until improvement or resolution, then reduce cabazitaxel dose from 25 mg/m ² to 20 mg/m ² .
Grade ≥ 2 peripheral neuropathy	Delay treatment until improvement, then reduce cabazitaxel dose from 25 mg/m ² to 20 mg/m ² .

The treatment should be discontinued if a patient continues to experience any of these reactions at 20 mg/m².

Special Populations

Patients with Hepatic Impairment

Cabazitaxel is extensively metabolised by the liver. No formal studies have been carried out in patients with hepatic impairment. As a precautionary measure, cabazitaxel should not be given to patients with hepatic impairment (bilirubin $\geq 1 \times$ Upper Limit of Normal (ULN), or AST and/or ALT $\geq 1.5 \times$ ULN) (see sections Contraindications, Special Warnings and Precautions for Use and Pharmacokinetic Properties).

Patients with Renal Impairment

Cabazitaxel is minimally excreted through the kidney. No dose adjustment is necessary in patients with mild renal impairment (creatinine clearance (CL_{CR}): 50 to 80 ml/min). Limited data are available for patients with moderate (CL_{CR}: 30 to 50 ml/min) and no data are available for patients with severe renal impairment (CL_{CR} < 30 ml/min) or end stage renal disease; therefore, these patients should be treated with caution and monitored carefully during treatment (see sections Special Warnings and Precautions for Use and Pharmacokinetic Properties).

Elderly

No specific dose adjustment for the use of cabazitaxel in elderly patients is recommended (see also sections *Special Warnings and Precautions for Use*, *Undesirable Effects* and *Pharmacokinetic Properties*).

Concomitant Medicinal Products Use

Concomitant medicinal products that are strong inducers or strong inhibitors of CYP3A should be avoided. However, if patients require co-administration of a strong CYP3A inhibitor, a 25% cabazitaxel dose reduction should be considered (see sections *Special Warnings and Precautions for Use* and *Interaction with Other Medicinal Products and Other Forms of Interaction*).

Paediatric Population

The safety and the efficacy of cabazitaxel in children and adolescents below 18 years of age have not been established. No data are available.

Method of Administration

For instructions on preparation and administration of the product, see section *Special Precautions for Disposal and Other Handling*.

PVC infusion containers and polyurethane infusion sets should not be used.

Cabazitaxel must not be mixed with any other medicinal products than those mentioned in section *Special Precautions for Disposal and Other Handling*.

8. Contraindications

- Hypersensitivity to cabazitaxel, to other taxanes, or polysorbate 80 or any excipients.
- Neutrophil counts less than 1,500/mm³.
- Hepatic impairment.

- Concomitant vaccination with yellow fever vaccine.

9. Special warnings and precautions for use:

Hypersensitivity reactions

All patients should be pre-medicated prior to the initiation of the infusion of cabazitaxel.

Patients should be observed closely for hypersensitivity reactions especially during the first and second infusions. Hypersensitivity reactions may occur within a few minutes following the initiation of the infusion of cabazitaxel, thus facilities and equipment for the treatment of hypotension and bronchospasm should be available. Severe reactions can occur and may include generalised rash/erythema, hypotension and bronchospasm. Severe hypersensitivity reactions require immediate discontinuation of cabazitaxel and appropriate therapy. Patients with a hypersensitivity reaction must stop treatment with cabazitaxel.

Bone marrow suppression

Bone marrow suppression manifested as neutropenia, anaemia, thrombocytopenia, or pancytopenia may occur. (See "Risk of neutropenia" and "Anaemia" in section warnings and precautions)

Risk of neutropenia

Patients treated with cabazitaxel may receive prophylactic G-CSF, as per American Society of Clinical Oncology (ASCO) guidelines and/or current institutional guidelines, to reduce the risk or manage neutropenia complications (febrile neutropenia, prolonged neutropenia or neutropenic infection). Primary prophylaxis with G-CSF should be considered in patients with high-risk clinical features (age >65 years, poor performance status, previous episodes of febrile neutropenia, extensive prior radiation ports, poor nutritional status, or other serious comorbidities) that predispose them to increased complications from prolonged neutropenia. The use of G-CSF has been shown to limit the incidence and severity of neutropenia.

Neutropenia is the most common adverse reaction of cabazitaxel. Monitoring of complete blood counts is essential on a weekly basis during cycle 1 and before each treatment cycle thereafter so that the dose can be adjusted, if needed.

The dose should be reduced in case of febrile neutropenia, or prolonged neutropenia despite appropriate treatment.

Patients should be re-treated only when neutrophils recover to a level $\geq 1,500/\text{mm}^3$.

Gastrointestinal disorders

Symptoms such as abdominal pain and tenderness, fever, persistent constipation, diarrhoea, with or without neutropenia, may be early manifestations of serious gastrointestinal toxicity and should be evaluated and treated promptly. Cabazitaxel treatment delay or discontinuation may be necessary.

Risk of nausea, vomiting, diarrhoea and dehydration

If patients experience diarrhoea following administration of cabazitaxel they may be treated with commonly used anti-diarrhoeal medicinal products. Appropriate measures should be taken to re-hydrate patients. Diarrhoea can occur more frequently in patients that have received prior abdomino-pelvic radiation. Dehydration is more common in patients aged 65 or older. Appropriate

measures should be taken to rehydrate patients and to monitor and correct serum electrolyte levels, particularly potassium. Treatment delay or dose reduction may be necessary for grade ≥ 3 diarrhoea. If patients experience nausea or vomiting, they may be treated with commonly used anti-emetics.

Risk of serious gastrointestinal reactions

Gastrointestinal (GI) hemorrhage and perforation, ileus, colitis, including fatal outcome, have been reported in patients treated with cabazitaxel. Caution is advised with treatment of patients most at risk of developing gastrointestinal complications: those with neutropenia, the elderly, concomitant use of NSAIDs, anti-platelet therapy or anti-coagulants, and patients with a prior history of pelvic radiotherapy or gastrointestinal disease, such as ulceration and GI bleeding.

Peripheral neuropathy

Cases of peripheral neuropathy, peripheral sensory neuropathy (e.g., paraesthesias, dysaesthesias) and peripheral motor neuropathy have been observed in patients receiving cabazitaxel. Patients under treatment with cabazitaxel should be advised to inform their doctor prior to continuing treatment if symptoms of neuropathy such as pain, burning, tingling, numbness, or weakness develop. Physicians should assess for the presence or worsening of neuropathy before each treatment. Treatment should be delayed until improvement of symptoms. The dose of cabazitaxel should be reduced from 25 mg/m² to 20 mg/m² for persistent grade ≥ 2 peripheral neuropathy.

Anaemia

Anaemia has been observed in patients receiving cabazitaxel. Haemoglobin and haematocrit should be checked before treatment with cabazitaxel and if patients exhibit signs or symptoms of anaemia or blood loss. Caution is recommended in patients with haemoglobin <10 g/dl and appropriate measures should be taken as clinically indicated.

Risk of renal failure

Renal disorders, have been reported in association with sepsis, severe dehydration due to diarrhoea, vomiting and obstructive uropathy. Renal failure including cases with fatal outcome has been observed. Appropriate measures should be taken to identify the cause and intensively treat the patients if this occurs.

Adequate hydration should be ensured throughout treatment with cabazitaxel. The patient should be advised to report any significant change in daily urinary volume immediately. Serum creatinine should be measured at baseline, with each blood count and whenever the patient reports a change in urinary output. Cabazitaxel treatment should be discontinued in case of any degradation of renal function to renal failure $\geq$ CTCAE 4.0 Grade 3.

Respiratory disorders

Interstitial pneumonia/pneumonitis and interstitial lung disease have been reported and may be associated with fatal outcome. (See section undesirable effects)

If new or worsening pulmonary symptoms develop, patients should be closely monitored, promptly investigated, and appropriately treated. Interruption of cabazitaxel therapy is recommended until diagnosis is available. Early use of supportive care measures may help improve the condition. The benefit of resuming cabazitaxel treatment must be carefully evaluated.

Risk of cardiac arrhythmias

Cardiac arrhythmias have been reported, most commonly tachycardia and atrial fibrillation.

Elderly

Elderly people (≥ 65 years of age) may be more likely to experience certain adverse reactions including neutropenia and febrile neutropenia.

Patients with liver impairment

Treatment with Cabazitaxel is contraindicated in patients with hepatic impairment.

Interactions

Co-administration with strong CYP3A inhibitors should be avoided since they may increase the plasma concentrations of cabazitaxel. If co-administration with a strong CYP3A inhibitor cannot be avoided, close monitoring for toxicity and a cabazitaxel dose reduction should be considered.

Co-administration with strong CYP3A inducers should be avoided since they may decrease plasma concentrations of cabazitaxel.

10. Fertility, pregnancy and lactation:

Pregnancy

There are no data from the use of cabazitaxel in pregnant women. Studies in animals have shown reproductive toxicity at maternotoxic doses and that cabazitaxel crosses the placenta barrier. As with other cytotoxic medicinal products, cabazitaxel may cause foetal harm in exposed pregnant women.

Cabazitaxel is not recommended during pregnancy and in women of childbearing potential not using contraception.

Breast-feeding

Available pharmacokinetics data in animals have shown excretion of cabazitaxel and its metabolites in milk. A risk to the suckling child cannot be excluded.

Cabazitaxel should not be used during breast-feeding.

Fertility

Animal studies showed that cabazitaxel affected reproductive system in male rats and dogs without any functional effect on fertility. Nevertheless, considering the pharmacological activity of taxanes, their genotoxic potential and effect of several compounds of this class on fertility in animal studies, effect on male fertility could not be excluded in human.

Due to potential effects on male gametes and to potential exposure via seminal liquid, men treated with cabazitaxel should use effective contraception throughout treatment and are recommended to continue this for up to 6 months after the last dose of cabazitaxel. Due to potential exposure via seminal liquid, men treated with cabazitaxel should prevent contact with the ejaculate by another person throughout treatment. Men being treated with cabazitaxel are advised to seek advice on conservation of sperm prior to treatment.

11. Drug interactions:

Cabazitaxel is mainly metabolised through CYP3A (80% to 90%).

CYP3A inhibitors

Repeated administration of ketoconazole (400 mg once daily), a strong CYP3A inhibitor, resulted in a 20% decrease in cabazitaxel clearance corresponding to a 25% increase in AUC. Therefore concomitant administration of strong CYP3A inhibitors (e.g., ketoconazole, itraconazole, clarithromycin, indinavir, nefazodone, nelfinavir, ritonavir, saquinavir, telithromycin, voriconazole) should be avoided as an increase of plasma concentrations of cabazitaxel may occur.

Concomitant administration of aprepitant, a moderate CYP3A inhibitor, had no effect on cabazitaxel clearance.

CYP3A inducers

Repeated administration of rifampin (600 mg once daily), a strong CYP3A inducer, resulted in an increase in cabazitaxel clearance of 21% corresponding to a decrease in AUC of 17%.

Therefore concomitant administration of strong CYP3A inducers (e.g., phenytoin, carbamazepine, rifampin, rifabutin, rifapentin, phenobarbital) should be avoided as a decrease of plasma concentrations of cabazitaxel may occur. In addition, patients should also refrain from taking St. John's Wort.

OATP1B1

Cabazitaxel has also been shown to inhibit the transport proteins of the Organic Anion Transport Polypeptides OATP1B1. The risk of interaction with OATP1B1 substrates (e.g. statins, valsartan, repaglinide) is possible, notably during the infusion duration (1 hour) and up to 20 minutes after the end of the infusion. A time interval of 12 hours is recommended before the infusion and at least 3 hours after the end of infusion before administering the OATP1B1 substrates.

Vaccinations

Administration of live or live-attenuated vaccines in patients immunocompromised by chemotherapeutic agents may result in serious or fatal infections. Vaccination with a live attenuated vaccine should be avoided in patients receiving cabazitaxel. Killed or inactivated vaccines may be administered; however, the response to such vaccines may be diminished.

12. Effects on ability to drive and use machines

Cabazitaxel may influence the ability to drive and use machines as it may cause fatigue and dizziness. Patients should be advised not to drive or use machines if they experience these adverse reactions during treatment.

13. Undesirable Effects

Summary of safety profile

The most commonly occurring adverse reactions in all grades were anemia, leukopenia, neutropenia, thrombocytopenia and diarrhea.

The most commonly occurring grade ≥ 3 adverse reactions in the cabazitaxel group were neutropenia, leukopenia, anemia, febrile neutropenia and diarrhea.

Discontinuation of treatment due to adverse reactions occurred in patients receiving cabazitaxel. The most common adverse reactions leading to cabazitaxel discontinuation was neutropenia.

Adverse reactions are listed according to MedDRA system organ class, adverse reactions and haematological abnormalities with cabazitaxel in combination with prednisone or prednisolone were shown below.

Infections and infestations:

Common: Septic shock, Sepsis, Cellulitis, Urinary tract infection, Influenza, Cystitis, Upper respiratory tract infection, Herpes zoster and Candidiasis.

Blood and lymphatic system disorders:

Very common: Neutropenia^{a*}, Anaemia^a, Leukopenia^a, Thrombocytopenia^a

Common: Febrile neutropenia

Immune system disorders:

Common: Hypersensitivity

Metabolism and nutrition disorders:

Very common: Anorexia

Common: Dehydration, Hyperglycaemia and Hypokalemia

Psychiatric disorders:

Common: Anxiety, Confusional state

Nervous system disorders:

Very common: Dysgeusia

Common: Neuropathy peripheral, peripheral sensory neuropathy, Dizziness, Headache, Paraesthesia, Lethargy, Hypoaesthesia and Sciatica.

Eye disorders

Common: Conjunctivitis: Lacrimation increased

Ear and labyrinth disorders:

Common: Tinnitus, Vertigo

Cardiac disorders:*

Common: Atrial fibrillation, Tachycardia

Vascular disorders:

Common: Hypotension, Deep vein thrombosis, Hypertension, Orthostatic hypotension, Hot flush, Flushing

Respiratory, thoracic and mediastinal disorders:

Very common: Dyspnoea, Cough

Common: Oropharyngeal pain, Pneumonia

Gastrointestinal disorders:

Very common: Diarrhoea, Nausea, Vomiting, Constipation, Abdominal pain

Common: Dyspepsia, Abdominal pain upper, Haemorrhoids, Gastroesophageal reflux disease, Rectal haemorrhage, Dry mouth, Abdominal distension

Skin and subcutaneous tissue disorders:

Very common: Alopecia

Common: Dry skin, Erythema

Musculoskeletal and connective tissue disorders:

Very common: Back pain, Arthralgia

Common: Pain in extremity, Muscle spasms, Myalgia, Musculoskeletal chest pain, Flank pain

Renal and urinary disorders:

Very common: Haematuria

Common: Acute renal failure, Renal failure, Dysuria, Renal colic, Pollakiuria, Hydronephrosis, Urinary retention, Urinary incontinence, Ureteric obstruction

Reproductive system and breast disorders:

Common: Pelvic pain

General disorders and administration site conditions:

Very common: Fatigue, Asthenia, Pyrexia

Common: Peripheral oedema, Mucosal inflammation, Pain, Chest pain, Oedema, Chills, Malaise

Investigations:

Common: Weight decreased, Aspartate aminotransferase increased, Transaminases increased.

a based on laboratory values

* see detailed section below

Description of selected adverse reactions

Neutropenia, and associated clinical events

Incidence of grade ≥ 3 neutropenia based on laboratory data were clinical neutropenia and febrile neutropenia. Neutropenia was the most common adverse reaction leading to medicinal product discontinuation.

Neutropenic complications included neutropenic infections, neutropenic sepsis, and septic shock, which in some cases resulted in a fatal outcome.

The use of G-CSF has been shown to limit the incidence and severity of neutropenia.

Cardiac disorders and arrhythmias:

All Grade events among cardiac disorders were more common on cabazitaxel of which patients had Grade ≥ 3 cardiac arrhythmias. Tachycardia and atrial fibrillation was reported in the cabazitaxel group. Cardiac failure events were more common on cabazitaxel and one patient died from cardiac failure. Fatal ventricular fibrillation, and cardiac arrest also seen in patients.

Haematuria

Haematuria all grades frequency has been reported. Confounding causes such as disease progression, instrumentation, infection or anticoagulation/NSAID/aspirin therapy were identified in nearly two thirds of the cases.

Other laboratory abnormalities:

The incidence of grade ≥ 3 were anaemia, increased AST, ALT, and increased bilirubin.

Gastrointestinal disorders

Colitis, enterocolitis, gastritis, neutropenic enterocolitis have been observed. Gastrointestinal hemorrhage and perforation, ileus and intestinal obstruction have also been reported.

Respiratory disorders

Cases of interstitial pneumonia/pneumonitis and interstitial lung disease, sometimes fatal have been reported with an unknown frequency (cannot be estimated from the available data).

Renal and urinary disorders

Cystitis due to radiation recall phenomenon, including haemorrhagic cystitis, were reported uncommonly.

Other special populations

Elderly population

The following adverse reactions reported at rates $\geq 5\%$ higher in patients 65 years of age or greater compared to younger patients: fatigue, clinical neutropenia, asthenia, pyrexia, dizziness, urinary tract infection and dehydration, respectively.

The incidence of the following grade ≥ 3 adverse reactions were higher in patients ≥ 65 years of age compared to younger patients; neutropenia based on laboratory abnormalities, clinical neutropenia and febrile neutropenia.

In prostate cancer patients, the adverse reactions reported at rates of at least 5% higher in patients 65 years of age or greater compared to younger patients were diarrhoea, fatigue, asthenia, constipation, clinical neutropenia, febrile neutropenia and dyspnea.

14. Overdosage

There is no known antidote to cabazitaxel. The anticipated complications of overdose would consist of exacerbation of adverse reactions as bone marrow suppression and gastrointestinal disorders.

In case of overdose, the patient should be kept in a specialized unit and closely monitored. Patients should receive therapeutic G-CSF as soon as possible after discovery of overdose. Other appropriate symptomatic measures should be taken.

15. Storage condition

Store below 30°C

Shelf life: 24 months from the date of manufacturing

Stability of the Second (Final) Dilution Solution In the Infusion Bag: Fully prepared cabazitaxel Injection In fusion solution (In either 0.9% sodium chloride solution or 5% dextrose solution) should be used within 8 Hours at ambient temperature (30°C) (including the one-hour Infusion), or for a total of 24 hours (Including the one-hour infusion) under the refrigerated conditions (2°C-8°C).

In addition, chemical and physical stability of the infusion solution has been demonstrated for 24 hours under refrigerated conditions (2°C-8°C). As both the first diluted solution and the second (final) infusion solution are supersaturated, the solutions may crystallize over time. If crystals and/or particulates appear, the solutions must not be used and should be discarded.

Precaution:

Cabazitaxel 60 mg/1.5 mL Concentrate and Solvent for Solution for Infusion is a cytotoxic anticancer drug. Follow applicable special handling and disposable procedures.

Keep out of reach of children.

Special precautions for disposal procedures and other handling

CABAZRED 60 mg/1.5mL Concentrate and Solvent for Solution for Infusion should only be prepared and administered by personnel trained in handling cytotoxic agents. Pregnant staff should not handle the product. As for any other antineoplastic agent, caution should be exercised when handling and preparing Cabazitaxel solutions, taking into account the use of containment devices, personal protective equipment (e.g. gloves), and preparation procedures. If CABAZRED 60 mg/1.5mL Concentrate and Solvent for Solution for Infusion, at any step of its handling, should come into contact with the skin, wash immediately and thoroughly with soap and water. If it should come into contact with mucous membranes, wash immediately and thoroughly with water.

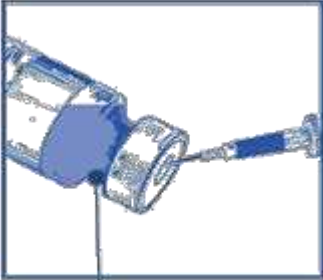
Always dilute the concentrate for solution for infusion with the **entire** supplied solvent before adding to infusion solution.

Read this **ENTIRE** section carefully before mixing and diluting. CABAZRED 60 mg/1.5mL Concentrate and Solvent for Solution for Infusion requires **TWO** dilutions prior to administration. Follow the preparation instructions provided below.

Note: Both the Cabazitaxel 60 mg/1.5 ml concentrate vial (fill volume: 73.2 mg of cabazitaxel/1.83 ml) and the solvent vial (fill volume: 5.7 ml) contain an overfill to compensate for liquid loss during preparation. This overfill ensures that after dilution with the **ENTIRE** contents of the accompanying solvent, there is solution containing 10 mg/ml cabazitaxel.

The following two-step dilution process must be carried out in an aseptic manner for preparing the solution for infusion.

Step 1: Initial dilution of the concentrate for solution for infusion with the supplied solvent.

Step 1.1 Inspect the concentrate vial and the supplied solvent. The concentrate solution and the solvent should be clear.	 Concentrate vial (60 mg - 1.5 ml) Solvent vial
Step 1.2 Using a syringe fitted with a needle, aseptically withdraw the entire contents of the supplied solvent by partially inverting the vial.	 Solvent vial
Step 1.3 Inject the entire contents into the corresponding concentrate vial. To limit foaming as much as possible when injecting the solvent, direct the needle onto the inside wall of the vial of concentrate solution and inject slowly. Once reconstituted, the resultant solution contains 10 mg/ml of cabazitaxel.	 Concentrate-solvent mixture 10 mg/ml Solvent vial
Step 1.4 Remove the syringe and needle and mix manually and gently by repeated inversions until obtaining a clear and homogeneous solution. It could take approximately 45 seconds.	 Concentrate-solvent mixture 10 mg/ml

Step 1.5

Let this solution stand for approximately 5 minutes and check then that the solution is homogeneous and clear.

It is normal for foam to persist after this time period.



Concentrate-solvent
mixture 10 mg/ml

This resulting concentrate-solvent mixture contains 10 mg/ml of cabazitaxel (at least 6 ml deliverable volume). The second dilution should be done immediately (within 1 hour) as detailed in Step 2.

More than one vial of the concentrate-solvent mixture may be necessary to administer the prescribed dose.

Step 2: Second (final) dilution for infusion

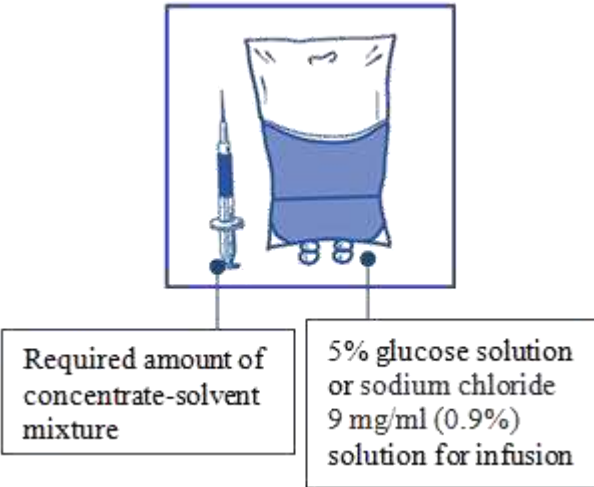
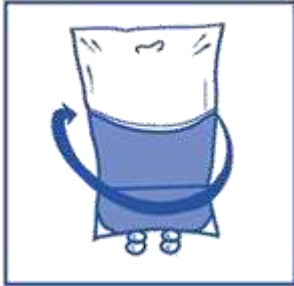
Step 2.1

Aseptically withdraw the required amount of concentrate-solvent mixture (10 mg/ml of cabazitaxel), with a graduated syringe fitted with a needle. As an example, a dose of 45 mg Cabazitaxel would require 4.5 ml of the concentrate-solvent mixture prepared following Step 1.

Since foam may persist on the wall of the vial of this solution, following its preparation described in Step 1, it is preferable to place the needle of the syringe in the middle when extracting.



Concentrate-solvent
mixture 10 mg/ml

Step 2.2 Inject in a sterile PVC-free container of either 5% glucose solution or sodium chloride 9 mg/ml (0.9%) solution for infusion. The concentration of the infusion solution should be between 0.10 mg/ml and 0.26 mg/ml.	
Step 2.3 Remove the syringe and mix the content of the infusion bag or bottle manually using a rocking motion.	
Step 2.4 As with all parenteral products, the resulting infusion solution should be visually inspected prior to use. As the infusion solution is supersaturated, it may crystallize over time. In this case, the solution must not be used and should be discarded.	

The infusion solution should be used immediately. However, in-use storage time can be longer under specific conditions mentioned in section 6.3.

An in-line filter of 0.22 micrometer nominal pore size (also referred to as 0.2 micrometer) is recommended during administration.

Do not use PVC infusion containers or polyurethane infusion sets for the preparation and administration of Cabazitaxel.

Cabazitaxel must not be mixed with any other medicinal products than those mentioned.

Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

16. Dosage forms and packaging available

Dosage forms:

Parenteral Preparation in form of Concentrates for Infusions

Presentation:

Cabazitaxel 60 mg/1.5 mL Concentrate and Solvent for Solution for Infusion is supplied as a kit consisting of the following:

- One single-dose vial of Cabazitaxel 60 mg/1.5 mL concentrate for Solution for Infusion: a clear yellow to brownish-yellow viscous solution free from visible extraneous matter in a clear glass vial with a rubber stopper and flip-off seal orange.
- One single-dose vial of Solvent for Cabazitaxel 60 mg/1.5 mL concentrate for Solution for Infusion: a clear colorless solution free from visible extraneous matter of 13% (w/w) ethanol in water for injection in a clear glass vial with a rubber stopper and flip-off seal mist grey.

Incompatibility:

This medicinal product must not be mixed with other medicinal products except those mentioned in posology and method of administration section.

PVC infusion containers or polyurethane infusion sets should not be used for the preparation and administration of the infusion solution.

17. Name and Address of Manufacturer

Manufactured in India by
Dr. Reddy's Laboratories Limited.,
Formulations Unit 7
Plot No. P1 to P9, Phase III
VSEZ, Duvvada,
Visakhapatnam District
Andhra Pradesh
INDIA.

For Further Information Write to:
Global Medical Affairs.
DR.REDDY'S LABORATORIES LTD,
8-2-337, Road No.3, Banjara Hills,
Hyderabad, India.

18. Date of Revision of Package Insert

December 27th, 2017